

Project Report

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Analyzing Space Weather using Data from different satellites

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INTRODUCTION

Out in the space, about 149.6 million km away from the Earth, is the center of our Solar System: the big ball of hot gases, the Sun. Even though it is at such a large distance from the Earth, its activity controls the life on Earth. Solar maxima and solar minima are the two extremes of the Sun's periodic 11-year activity cycle, called solar-cycle. ^[1] During Solar maxima, the Sun is highly active and a rise in the level of Coronal Mass Ejections (CMEs) and number of Sunspots and Solar flares is observed. Emission of Solar Energetic Particles (SEPs) is a major consequence of CMEs and solar flares.

Just like a bar-magnet, the Earth's magnetic field also has a North Pole and a South Pole. It prevents the entry of most of the charged particles from the Sun, but at higher latitudes these particles are guided by the field lines towards the poles, leading to the formation of auroras- northern and southern lights ^[2]. Also, the magnetic field lines exert a force on the charged particles, giving rise to the formation of radiation belts. A radiation belt is a region of energetic charged particles, trapped due to a planet's magnetic field ^[3]. The Earth's radiation belts are called Van Allen Radiation belts, named after the person who discovered them-James Van Allen. Earth has two such belts: Inner Radiation Belt and Outer Radiation Belt. The South Atlantic Anomaly (SAA) is the region where the Earth's inner Van Allen radiation belt comes closest to the Earth's surface, down to an altitude of 200 km. ^[4]

When a CME reaches the Earth, it disturbs the magnetic field of the Earth causing a geomagnetic storm. A major threat is to the satellites orbiting the Earth at higher altitudes. The charged particles may highly charge the satellites, and a component may be damaged due to high current discharging into the satellite. Also, it is dangerous for astronauts on mission to Mars or Moon to be exposed to such radiation of high energy. Hence, it becomes important to study the Space Weather. ^[5]

OBJECTIVE: The main objective is to understand the Space Environment using data from radiation detectors onboard various satellites, namely PROBA-1, INTEGRAL and GOES-10. 1. All data processing was performed off-line using a commercial software package (MATLAB 2016a, The MathWorks Inc., Natick, MA, 2000).

RADIATION INTERACTION

In order to analyze the data from the detectors, it is important to first understand the interaction of the charged particles with the material of the detector.^[6]

I. Heavy Charged Particles

The heavy charged particles (including protons, α -particles, etc.) interact mostly with the electrons of the detector atoms (interactions with the nuclei are rare). The interaction may result in either the excitation or the ionization of the atoms, depending upon the proximity of encounter between the two. The charged particle transfers its energy to the electrons resulting in ion-pairs formation. Some detectors are based on suppressing the recombination of these ion-pairs to form neutral atoms.

As the particle traverses the detector, it loses energy and its track is close to straight line except at the end. The linear Stopping power is defined as the ratio differential energy loss for that particle within the absorber medium to the corresponding path length, or,

$$S = - \frac{dE}{dx}$$

The Stopping power is also called specific energy loss or “rate” of loss.

It is given by the Bethe formula,

$$\frac{-dE}{dx} = \frac{4\pi e^4 z^2}{m_0 v^2} NB \quad \text{--- (1)}$$

where

$$B = Z \left[\ln \frac{2m_0 v^2}{I} - \ln \left(1 - \frac{v^2}{c^2} \right) - \frac{v^2}{c^2} \right] \quad \text{--- (2)}$$

v and ze : velocity and charge of primary particle

N : number density of absorber atoms

Z : atomic number of absorber atoms

m_0 : electron rest mass

I : average excitation and ionization potential of the absorber

Only the first term in B is of importance for non-relativistic particles ($v \ll c$). The expression for B varies slowly with particle energy and hence the multiplicative factors assist in generalizing the behavior of the stopping power, S . Hence, for non-relativistic particles, S varies inversely with energy of the particle, $1/v^2$. One can account for this as- when the velocity of the particle is low, it spends a greater time in the vicinity of the electrons of the absorber atoms and as a result the impulse imparted to the electron and hence the transfer of energy by the particle is largest. For different particles having same initial velocity, the particle with the greatest charge will have the highest value for specific energy loss. As an example, specific energy loss of α -particles would be more than that of protons, of same initial energy, in a given detector medium. For comparing different materials, the term NZ is taken into consideration, and thus, materials with higher atomic number result in higher specific energy loss.

At low particle energies, the Bethe formula becomes invalid for the exchange of charges between the particles and the atoms of the absorber has to be taken into consideration.

II. Fast Electrons

The electrons too interact with the electrons of the absorber atoms but now, it can lose much more energy while interacting with the electrons of the detector, for the masses of both are equal. As a result, the track of electron is far away from straight lines, opposite to the case of heavy charged particles.

Expression for fast electrons, similar to (1), has also been given by Bethe to describe the stopping power due to collisional losses (ionization and excitation)

$$-\left(\frac{dE}{dx}\right)_c = \frac{2\pi e^4 N Z}{m_0 v^2} \left(\ln \frac{m_0 v^2 E}{2 I^2 (1 - \beta^2)} - (\ln 2)(2 \sqrt{1 - \beta^2} - 1 + \beta^2) + (1 - \beta^2) + \frac{1}{8} (1 - \sqrt{1 - \beta^2})^2 \right) \text{----- (3)}$$

where $\beta=v/c$ and other variables have the same meaning as in (1).

The energy loss characteristics of electrons differ from that of heavy charged particles for electrons also lose energy through radiative processes: bremsstrahlung radiation. The deflections in the path of electrons in interaction with the detector atoms are responsible for these losses. And the formula for specific energy loss due to these radiative processes is given by:

$$-\left(\frac{dE}{dx}\right)_r = \frac{NEZ(Z+1)e^4}{137 m_o^2 c^4} \left(4 \ln \frac{2E}{m_o c^2} - \frac{4}{3} \right) \text{----- (4)}$$

The heavy charged particles do not lose energy through radiative processes, as can be understood from the m_o^2 factor in the denominator of the above expression. And, clearly energy loss through radiative process is more for particles of higher energies (E) and for detector materials of higher atomic number (Z^2).

The total linear stopping power for electrons is given by:

$$\frac{dE}{dx} = \left(\frac{dE}{dx}\right)_c + \left(\frac{dE}{dx}\right)_r \text{----- (5)}$$

The next section will discuss about the radiation detectors that are based on these interactions.

RADIATION DETECTOR

By understanding the behavior of the radiation in different materials, different detectors are built. The net result of radiation interaction for most of the detectors is the appearance of a given amount of electric charge within the detector active volume. A detector collects this charge to form a basic signal. To do so, an electric field is imposed within the detector that causes the positive and negative charged particles to move in opposite directions.

There are three general modes of operation of radiation detection- pulse mode, current mode and mean square voltage (MSV) mode. All three are operationally distinct but they all use the same basic signal of collection of charge accumulated in the detector. At very high event rates, current mode or MSV mode is used. In current mode, the current recorded by the detector in fixed response time of the detector, T is time-averaged over T . This averaged current depends on the interaction rate and charge per interaction. But due to random fluctuations, there is statistical uncertainty in obtaining constant averaged current. Assuming charge produced in each event is constant and using Poisson statistical distribution, it turns out that this deviation is proportional to charge per interaction. And, it is this deviation that is squared in MSV mode, making the signal proportional to square of the charge produced per interaction. The MSV mode is most useful when measurements are being made in mixed radiation environments when the charge produced by one type of radiation is very slightly different from the second type. In pulse voltage mode, the output of a detector consists of a sequence of individual signal pulses, amplitude of each representing the results of single quantum of radiation interaction within the detector. Numbers of pulses above amplitude 0 to max amplitude are registered by a counting circuit and graphs are plotted between number of pulses exceeding the amplitude H and pulse height, H that assist in analysing the radiation interaction. The differential curve of this plot is most commonly referred for analysis by radiation instrumentalists. The minima of the differential curve are referred to as counting plateaus, which represents the areas of operation in which minimum sensitivity to drifts in discrimination levels are achieved. The detectors work in PSV mode when the countrates are not very high.

STANDARD RADIATION ENVIRONMENT MONITOR

a) Technological Details

The Standard Radiation Environment Monitor (SREM) is an instrument (developed by Contraves Space A.G. in collaboration with the Paul Scherrer Institute for Astrophysics), on board various European Space Agency (ESA) satellites to detect high-energy protons (from ~ 10 MeV to ~ 300 MeV) and electrons (~ 300 keV to ~ 6 MeV) and provides information about their energy spectrum with $(+/-)20^\circ$ angular resolution.^[7]

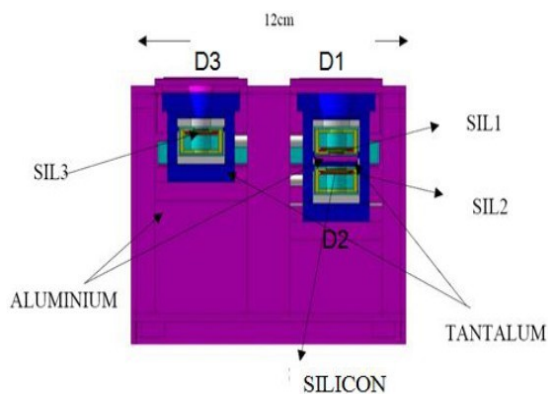


Fig. 1: SREM^[8]

The SREM is contained in a single box of $20 \times 12 \times 10$ cm³ and weighs 2.6 kg. It has three solid state silicon detectors (D1, D2 and D3) in two detector-heads configuration (Fig. 1). The detector D3 is a single detector and has an entrance window covered with 0.7 aluminium which sets the lower energy threshold for electrons ~ 0.5 MeV and for protons ~ 10 MeV.

The other two detectors D1 and D2 are arranged in telescope configuration. This system has an entrance window of 2mm thick layer of aluminium giving rise to proton threshold of 20 MeV and electron threshold of 1.5 MeV. The two detectors are separated by 1.7 mm thick layer of aluminium and 0.7 mm thick layer of tantalum that allows no electrons to pass through D2 and only protons above 43 MeV go through. Hence, the advantage of using two detectors in coincidence is that it gives measurement of pure proton countrates and also allows the subtraction of proton contamination from electron channels.

There are total 15 discriminator levels available with SREM to bin the energy of the particles crossing the detector. To raise an alarm flag, when the countrates exceed a programmable threshold, any two of the levels can be used.

The data was analysed of two different satellites: PROBA-1 and INTEGRAL supported by European Space Agency (ESA). Both the spacecraft have SREM onboard them. The

SREM onboard INTEGRAL has been adapted from the standard SREM configuration and is thus referred to as INTEGRAL/IREM. Table 1 gives some information relating to the position for both the satellites.

Table 1 Details about the three satellites

Info\Sat.	PROBA-I	INTEGRAL
Orbit	Low Earth Orbit (Sun-Synchronous)	High Earth Orbit (Highly-eccentric)
Inclination(deg.)	97.9	51.6
Time Period	97 min	72 hours
Instrument	SREM	IREM

The advantage of using data for two different satellites is that it would help in studying the environment in different regions of the magnetosphere and it will also support cross-calibration activities.

The counters for SREM and IREM along with the ranges of energy to which they are sensitive to, for protons and electrons are given in Table 2 and Table 3 respectively.^[8] The SCF represents the scaling factor that allows converting detector countrates to incident particle fluxes. And “n/a” indicates not available. The counters C1, C2, and C3 are coincident proton channels and hence are not sensitive to electrons.^[9]

Table 2 Counters for SREM onboard PROBA-1

No .	Counter	Detector	Proton Energy (MeV)			Electron Energy (MeV)		
			Min.	Max.	SCF(#/cm ²)	Min.	Max.	SCF(#/cm ²)
1	TC1	D1	26.00	∞	21.15	2.00	∞	169.01
2	S12	D1	26.00	∞	21.28	2.03	∞	361.09
3	S13	D1	26.00	∞	22.22	2.13	∞	1053.33
4	S14	D1	24.00	566.00	43.79	2.73	∞	86026.21
5	S15	D1	22.00	646.00	88.26	8.18	∞	8451.06
6	TC2	D2	49.00	∞	16.62	3.10	∞	174.40
7	S25	D2	53.00	318.00	209.40	n/a	n/a	n/a
8	C1	D1xD2	42.00	114.00	128.97	n/a	n/a	n/a
9	C2	D1xD2	51.00	270.00	91.42	n/a	n/a	n/a

10	C3	D1xD2	73.00	418.00	51.75	n/a	n/a	n/a
11	C4	D1xD2	154.00	∞	17.03	7.75	∞	216.37
12	TC3	D3	12.00	∞	56.63	0.80	∞	203.36
13	S32	D3	12.00	∞	56.87	0.73	∞	438.66
14	S33	D3	12.00	∞	57.35	1.03	∞	2198.48
15	S34	D3	11.00	∞	86.13	1.98	∞	171428.90

Table 3 Counters for IREM onboard INTEGRAL

No	Counter	Detector	Proton Energy (MeV)			Electron Energy (MeV)		
			Min.	Max.	SCF(#/cm ²)	Min.	Max.	SCF(#/cm ²)
1	TC1	D1	27.00	∞	15.83	2.00	∞	117.96
2	S12	D1	26.00	∞	19.00	2.08	∞	194.62
3	S13	D1	27.00	∞	16.00	2.23	∞	518.92
4	S14	D1	24.00	566.00	38.49	3.20	∞	25402.99
5	S15	D1	23.00	646.00	65.62	8.18	∞	5460.37
6	TC2	D2	49.00	∞	13.10	2.80	∞	191.46
7	S25	D2	48.00	318.00	208.80	n/a	n/a	n/a
8	C1	D1xD2	43.00	114.00	107.22	n/a	n/a	n/a
9	C2	D1xD2	52.00	270.00	75.63	n/a	n/a	n/a
10	C3	D1xD2	76.00	418.00	35.14	n/a	n/a	n/a
11	C4	D1xD2	164.00	∞	10.42	8.10	∞	154.61
12	TC3	D3	12.00	∞	49.29	0.80	∞	100.72
13	S32	D3	12.00	∞	49.34	0.75	∞	189.47
14	S33	D3	12.00	∞	40.17	1.05	∞	1162.18
15	S34	D3	12.00	∞	63.78	2.08	∞	93077.70

b) Data Analysis

The data from PROBA-1 and INTEGRAL is available online in CDF (Common Data format^[11]).^[10] The files are named “SREMPROBA1_PACC_yyyymmdd.cdf” and “IREM_PACC_yyyymmdd.cdf”, and contain deadtime corrected countrates and auxiliary data.^[12] The main variables contained in .cdf files are listed in Table 4.

Table 4 Information stored in .cdf files from IREM and SREM

Variable	No. of Elements	Description
EPOCH	1	Time since 0 AD of center of accumulation interval in ms
label_COUNTERS	15	Names of the counters, e.g. tc1, tc2, etc.
COUNTRATES	15	Deadtime corrected countrates corresponding to each counter
label_ORBIT	3	Labels for Cartesian components of spacecraft's location
ORBIT	3	Values for Cartesian components describing spacecraft's location in ECI system at given EPOCH
label_MAGFIELD	3	Labels for three components of computed magnetic field
MAGFIELD	3	Components of magnetic field at location of spacecraft at given EPOCH
LSHELL	1	Computed value of McIlwains L-shell parameter

There are other variables present in the file that give information about energy spectrum for electrons and protons at given EPOCH.

To work with .cdf files, the software- 'Matlab 2016a' was used. There are various functions and one package available in the software's library to handle .cdf files as mentioned in Table 5.

Table 5 Functions and Packages available in Matlab to work with .cdf files

Functions

cdfinfo	Gives information about Common Data Format (CDF) file
cdfread	Helps in reading data from Common Data Format (CDF) file
cdfepoch	Used to convert date string or serial date number to CDF formatted dates
todatenum	Converts CDF epoch object to MATLAB serial date number

Packages

cdflib	Allows to interact directly with CDF library
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A simple programming code was written to read a .cdf file and extract all the values of the variables into Matlab's workspace, allowing easy handling of data.

The countrates for two different counters: TC1 and S14 were plotted with time as shown in Fig. 2 and Fig. 3.

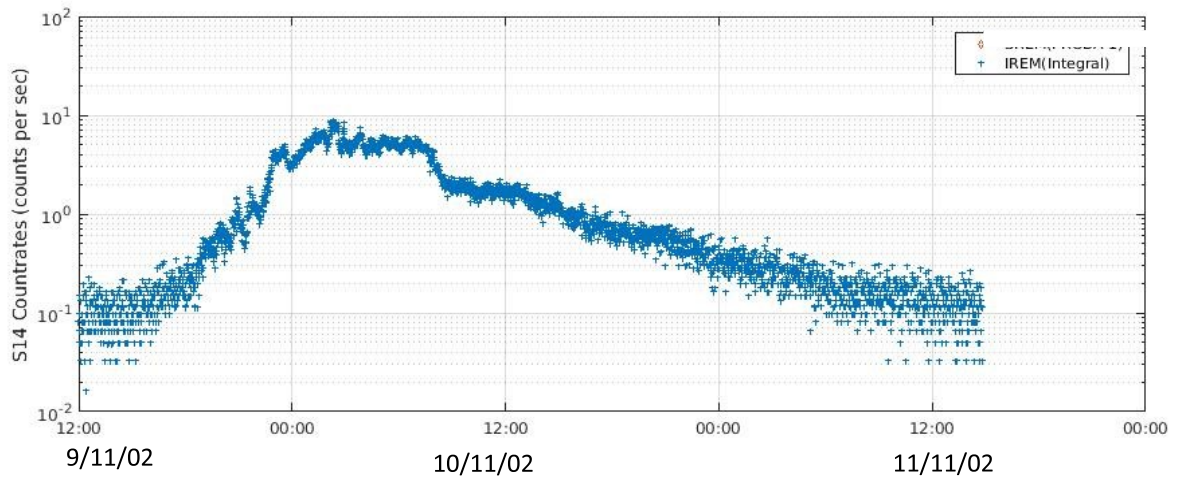


Fig. 2: Countrates for 9 and 11 Nov 2002 for S14 counter of IREM

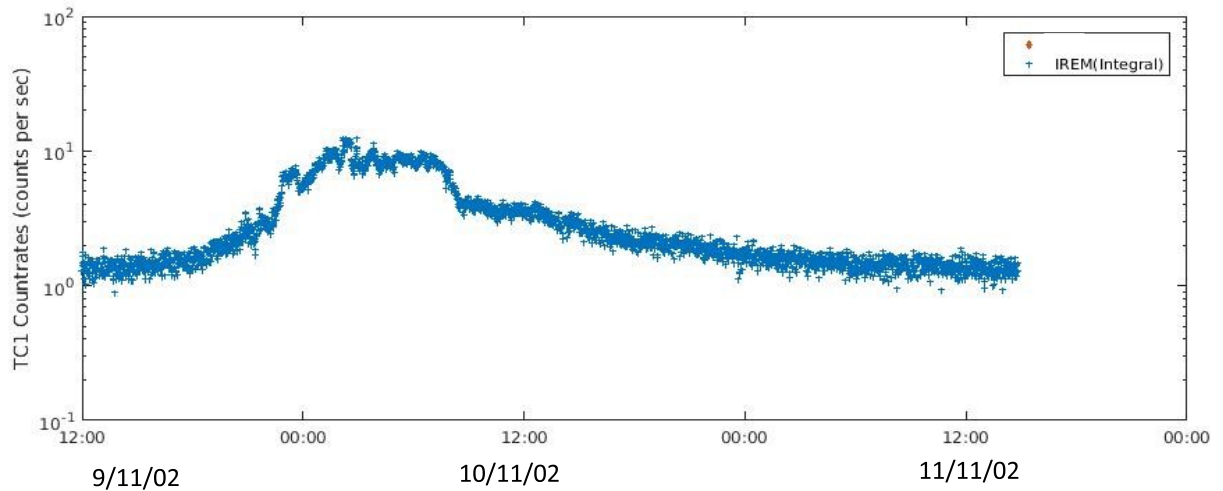


Fig. 3: Countrates for 9 and 11 Nov 2002 for TC1 counter of IREM

The spacecraft, INTEGRAL spends 70% of its time period in a region outside the radiation belts, in the interplanetary space and is not significantly shielded by the Earth's magnetosphere, thus measuring the radiation environment (i.e. solar protons). Fig.2 and Fig. 3 give information about a small solar proton event that was observed during the period of 9 Nov to 11 Nov 2002. The countrates started to increase from the end of 9 Nov until a plateau was reached in the morning of 10 Nov, and the proton environment was maintained till the end of 10 Nov. And, slowly the countrates started to decrease, reaching the quiescent level on 11 Nov. During this entire period, the spacecraft was not subjected to significant geomagnetic shielding.

For comparison, the countrates for S14 counter were plotted for SREM (PROBA-1) and were superimposed with that of IREM as shown in Fig. 4. The satellite PROBA-1 is in Low Earth Orbit and hence there are fluctuations. Whenever it is subjected to geomagnetic shielding the countrates are low and whenever it crosses the radiation belts, the countrates become significantly high. This indicates that the Earth's magnetosphere shields us from the charged particles and also complements the presence of radiation belts. When the countrates were plotted only for LSHELL values ≥ 8 , the profiles were closely matching as shown in Fig. 5.

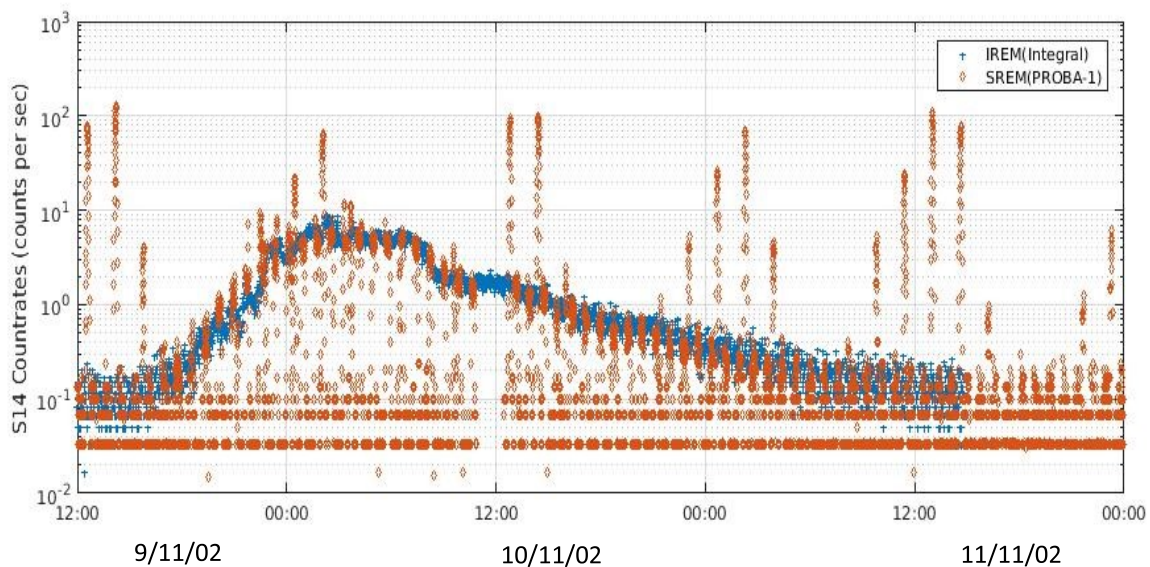


Fig. 4: Countrates of SREM superimposed with that of IREM for 9 Nov to 11 Nov 2002

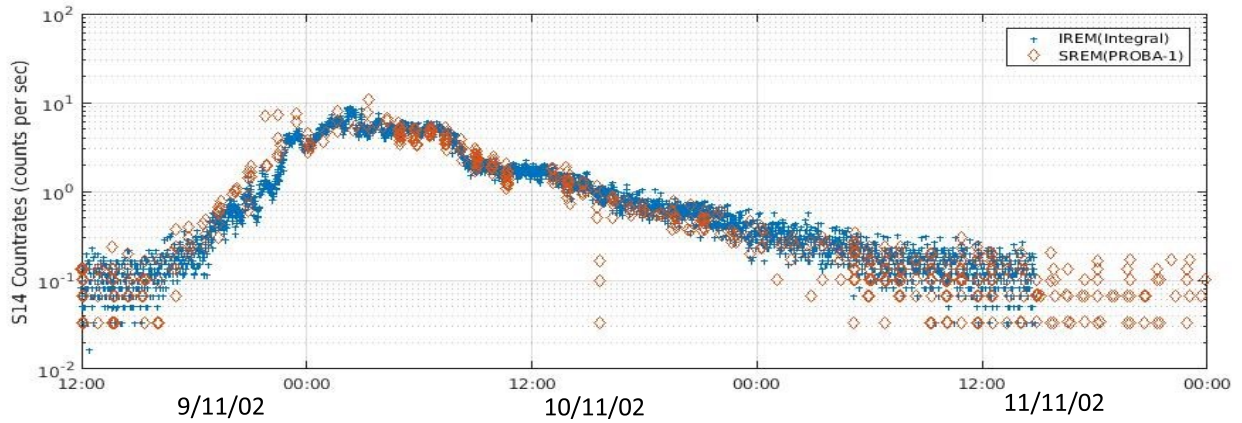


Fig. 5: Countrates of SREM filtered by plotting only for $LSHELL \geq 8$

Next, to understand the particle environment with respect to position of the Earth, the ECI coordinates at a given epoch were converted to geocentric latitude and longitude values.^[13] Using the lat-long values, the countrates were scattered on a world map as shown in Fig. 6 and Fig. 7. Fig. 6 describes the countrates of S14 channel for quiescent period of 6 and 7 Nov 2002. Notice the familiar high countrates observed in the South Atlantic Anomaly (SAA) region. The same was done for the small solar proton event (Fig. 7). One can compare both the figures and observe lighter shades of colors in the polar horn region indicating high countrates and complementing the solar proton event. The same can also be plotted for INTEGRAL as shown in Fig. 8.

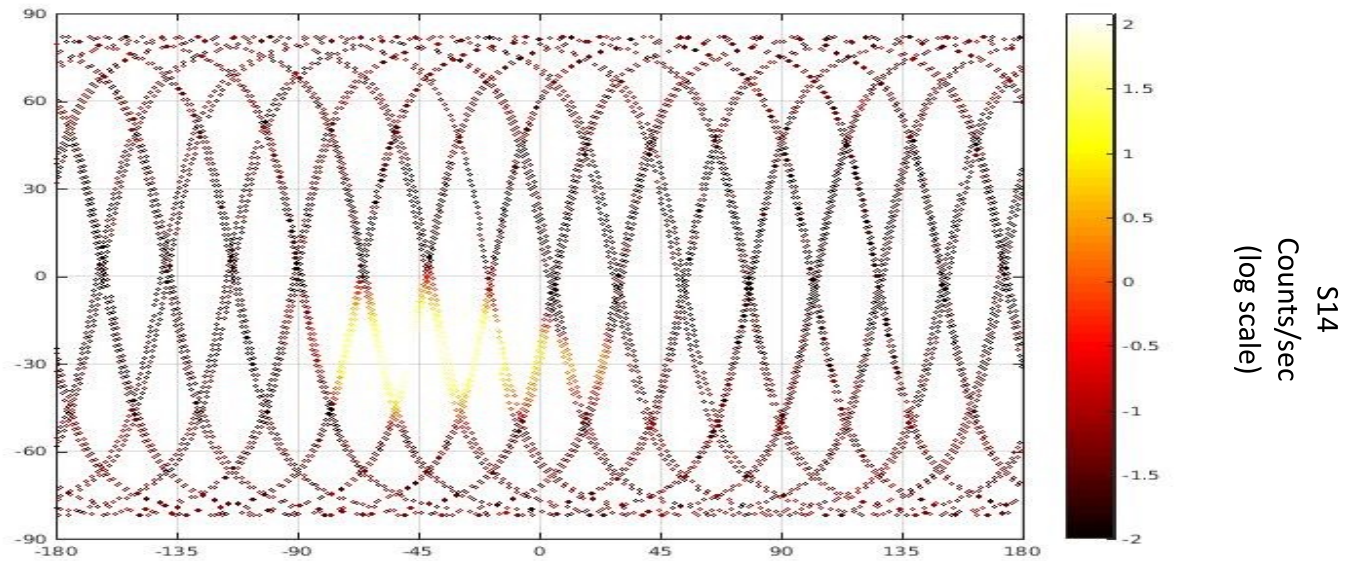


Fig. 6: Count rates of S14 counter plotted on a World-map for 6 & 7 Nov 2002 for SREM

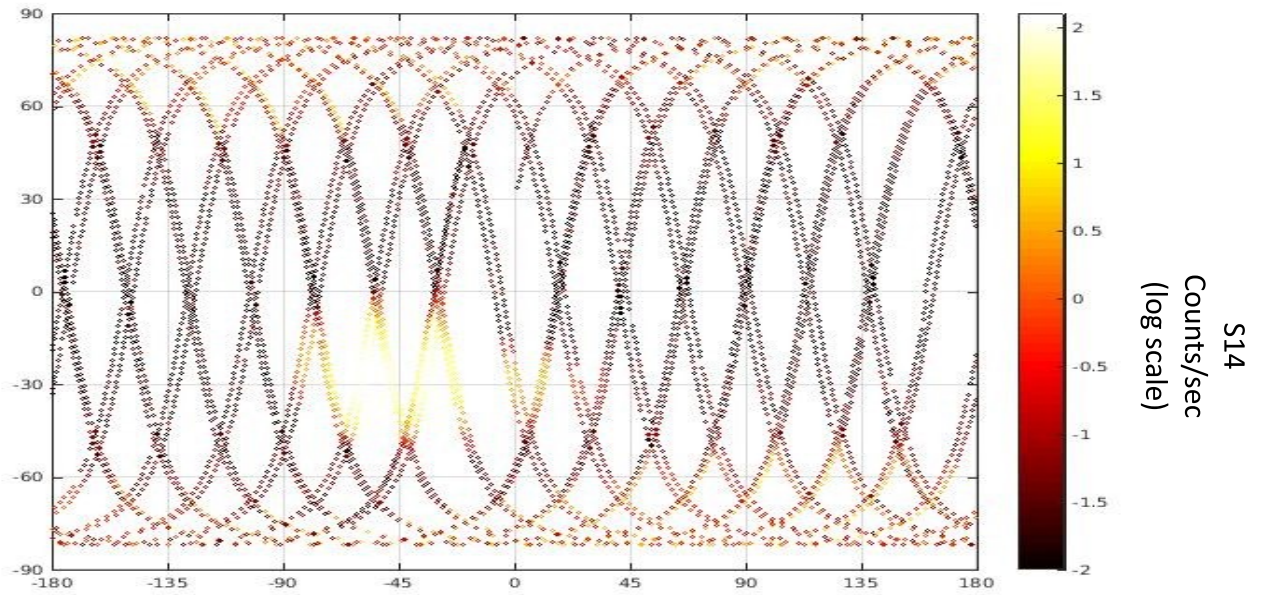


Fig. 7: Count rates of S14 counter plotted on a World-map for solar proton event of 9 to 11 Nov 2002 for SREM

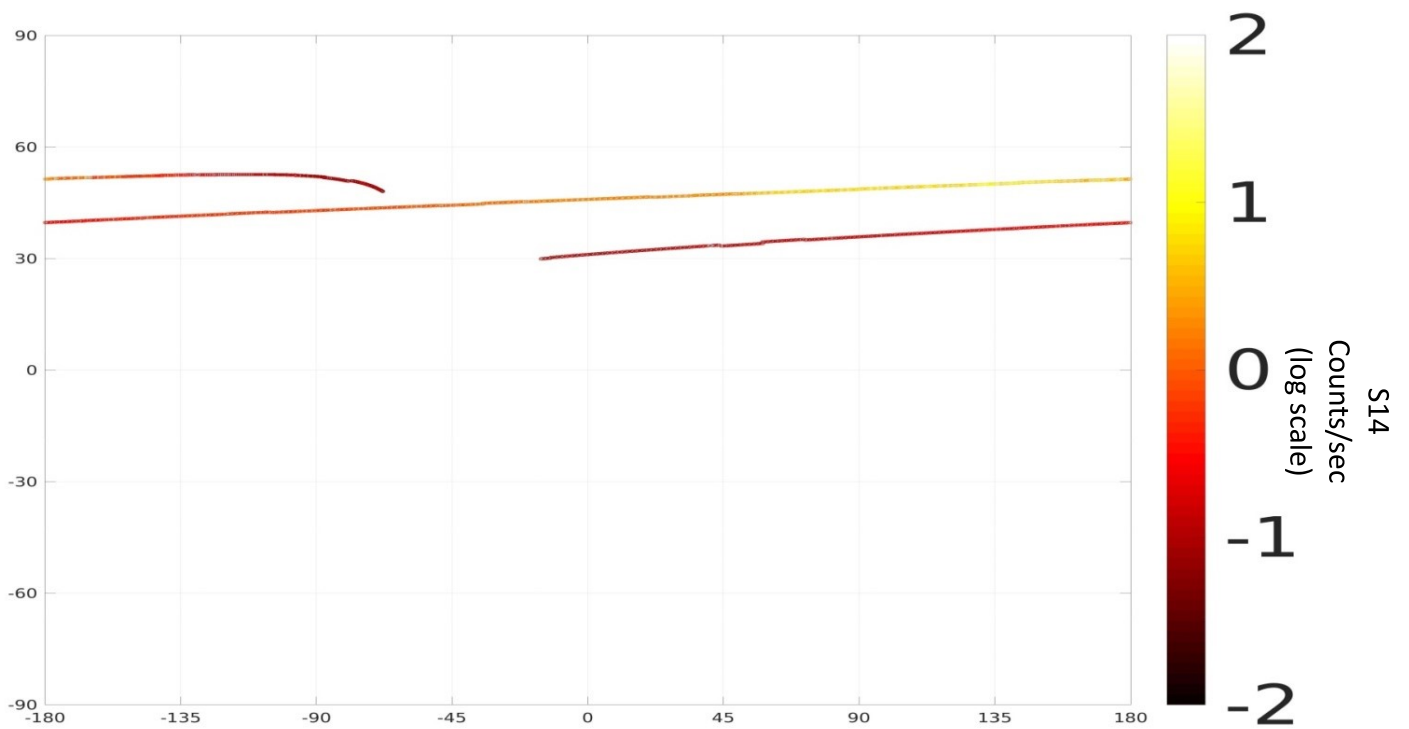


Fig. 8: Count rates of S14 counter plotted on a World-map for solar proton event of 9 to 11 Nov 2002 for IREM

SPACE ENVIRONMENT MONITOR

a) Technological Details

The Space Environment Monitor (SEM) onboard NASA's satellite: GOES-10, like SREM, also monitors the space environment. The satellite is geostationary and is positioned at 135°W and the equator. The SEM has four instruments namely, X-Ray Sensor (XRS), Magnetometers, High Energy Proton and Alpha Detector (HEPAD) and Energetic Particle Sensor (EPS). They give information about X-ray photon fluxes, magnetic field for the spacecraft and energetic particle fluxes.^[14]

X-Ray Sensor

The XRS is an X-ray telescope that provides in real-time solar X-ray flux in the spectral range of 0.5 to 4 angstroms (short sun channel) and 1 to 8 angstroms (long sun channel). X-rays are detected by two ion chambers, one for each spectral range.

Magnetometers

There are two three-axis magnetometers onboard GOES-10 that provide redundancy for measuring the geomagnetic field. They have a resolution of 0.03 nanoTesla.

HEPAD

The HEPAD measures incident flux of high energy protons (350 to greater than 700 MeV) and alphas (640 to greater than 850 MeV/nucleon). It has a telescope subassembly with two silicon surface barrier detectors, a Cerenkov radiator, and a photomultiplier tube (PMT), all arranged in a telescope configuration, and a signal analyzer subassembly. The solid-state detectors differentiate between minimum ionizing protons and alpha particles and aluminum and tungsten barriers shield from protons below 70 MeV and electrons below 15 MeV.

Energetic Particle Sensor (EPS)

The EPS allows three integral measurements of electrons from 0.6 to more than 4.0 megaelectronvolt (MeV), a seven-channel differential analysis of protons from 0.8 to 500 MeV, and a six-channel differential analysis of alpha particles from 4 to 500 MeV per nucleon, as given in Table 6. It also provides all the assistance required by the

HEPAD, which extends the energy ranges to greater than 700 MeV for protons and to greater than 3400 MeV per nucleon for alphas. The EPS consists of detectors in telescope assembly and dome assembly.

Table 6 Energy Ranges for EPS and HEPAD

Particle Type	Channel designation	Nominal Energy Range (MeV)	Detector Assembly
Proton	P1	≤ 0.8 to 4	Telescope
Proton	P2	4 to 9	Telescope
Proton	P3	9 to 5	Telescope
Proton	P4	15 to 40	Dome
Proton	P5	40 to 80	Dome
Proton	P6	80 to 165	Dome
Proton	P7	165 to 500	Dome
Proton	P8	350 to 420	HEPAD
Proton	P9	420 to 510	HEPAD
Proton	P10	510 to 700	HEPAD
Proton	P11	> 700	HEPAD
Alpha	A1	4 to 10	Telescope
Alpha	A2	10 to 21	Telescope
Alpha	A3	21 to 60	Telescope
Alpha	A4	60 to 150	Dome
Alpha	A5	150 to 250	Dome
Alpha	A6	300 to 500	Dome
Alpha	A7	2560 to 3400	HEPAD
Alpha	A8	> 3400	HEPAD
Electron	E1	≥ 0.6	Dome
Electron	E2	≥ 2.0	Dome
Electron	E3	≥ 4.0	Dome
“Singles”	S1 to S5	—	HEPAD

b) Data Analysis

The data files from GOES-10 are available online in two formats: .cdf and .txt.^[15] For analyzing both types of files, again, Matlab 2016a was used. In case of text files, .zip files are available and on unzipping them, many text files are obtained, each providing

with the information about one particular variable. The table 7 describes the data received from .txt files. On the other hand, the .cdf files mainly provide info about the spacecraft's location in different coordinates at the given epoch. Table 8 describes the data variables obtained from .cdf files.

Table 7 Information about data stored in .txt files

Information	Units	Range
Integral Electron Counts	particles/cm ² -s-sr	E1,E2,E3
Magnetic Field (3 axis and total)	nanoTesla	+512nT
Solar X-ray Flux	W/m ²	(0.5 to 4Å) & (1 to 8Å)
Corrected Integral Proton Counts	particles/cm ² -s-sr	>1,5,10,30,50,60,100 MeV
HEPAD Counts	particles/cm ² -s-sr	P8 to P11, A7, A8
Diff. Proton Counts	particles/cm ² -s-sr/MeV	P1 to P7

Table 8 Information about data stored in .cdf files

Variable	Description
'Epoch'	'Time'
'Time_PB5'	'Time'
'E1'	'GOES Electron Flux (> 0.6 MeV)'
'E2'	'GOES Electron Flux (> 2 MeV)'
'E3'	'GOES Electron Flux (> 4 MeV)'
'P1'	'GOES Uncorrected. Diff. Proton Flux (0.7-4 MeV)'
'SC_pos_ll'	'GOES 10 position, Geographic'
'SC_pos_eo'	'GOES 10 position, GEO'
'SC_pos_se'	'GOES 10 position, GSE'
'SC_pos_sm'	'GOES 10 position, GSM'
'EPS_stat'	'0=OK, 1=minor problems, 2=major problems, 3=mis...'
'label_time'	'Label for Time_PB5'
'unit_time'	'Units for Time_PB5'
'format_time'	'Format for Time_PB5'
'E1_flag'	'Post-gap Flag: 0-no gap, 1-data n/a'
'E2_flag'	'Post-gap Flag: 0-no gap, 1-data n/a'
'E3_flag'	'Post-gap Flag: 0-no gap, 1-data n/a'
'P1_flag'	'Post-gap Flag: 0-no gap, 1-data n/a'

'cartesian'	'Label, S/C Position'
'SC_flag'	'GOES 10 Post-gap Flag: 0-no gap, 1-data n/a'
'label_latlong'	'GOES 10 Position, Geographic'
'label_GEO'	'GOES 10 Position, GEO'
'label_GSE'	'GOES 10 Position, GSE'
'label_GSM'	'GOES 10 Position, GSM'
'unit_sc'	'GOES 10 Position, Geographic units'

Comparison b/w IREM, SREM and SEM

To compare data from the three satellites, the countrates from IREM and SREM had to be converted into particle fluxes having units same as that of SEM- counts/cm²-s-sr-MeV. For cm², the scaling factors, SCF were used and for MeV, the countrates were divided by the range of the energy the counter is sensitive to, and for steradian (sr), the countrates were divided by 4pi, assuming the omnidirectional countrates were provided by SREM and IREM.

The proton fluxes for SEM, IREM and SREM were plotted (Fig.9) for the Halloween Solar Storm- a powerful solar storm observed during the period of mid Oct to early Nov 2003 (Solar Cycle 23) that had some serious consequences. As an example, the Solar and Heliospheric Observatory (SOHO) satellite which is collaboration between NASA and the European Space Agency (ESA), failed temporarily.^[16] Clearly, the profiles for all three satellites are similar for the storm. And again, SREM displays fluctuations for PROBA-1 steps in and out of the region of geomagnetic shielding and radiation belts. The data for PROBA-1 was not available for all the dates and hence there is a gap in between. To further strengthen the analysis, the solar x-ray fluxes obtained from X-Ray Sensor of SEM, were also plotted on the same graph, Fig. 10. It can be observed that sometimes a peak in the x-ray fluxes also corresponds to a peak in particle fluxes indicating the occurrence of Solar flare and Coronal Mass Ejections at the same time, also, the fact that the particles reaching the detectors were relativistic. Sometimes, a peak in particle fluxes is observed after a peak in x-ray fluxes implying a lag between Solar flare and CME. The same can also be observed in Fig. 11, where the yellow lines represent the peak time for solar flares obtained from Hessi Solar Flare list provided by NASA.^[17]

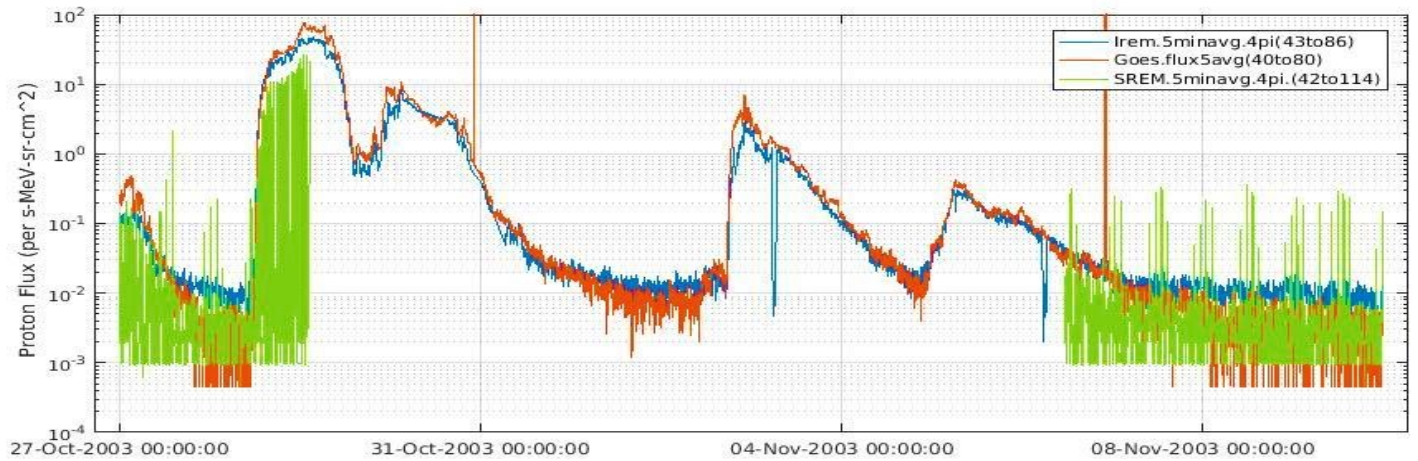


Fig. 9: Incident Proton Fluxes for Halloween Solar Storm for IREM, SREM & SEM

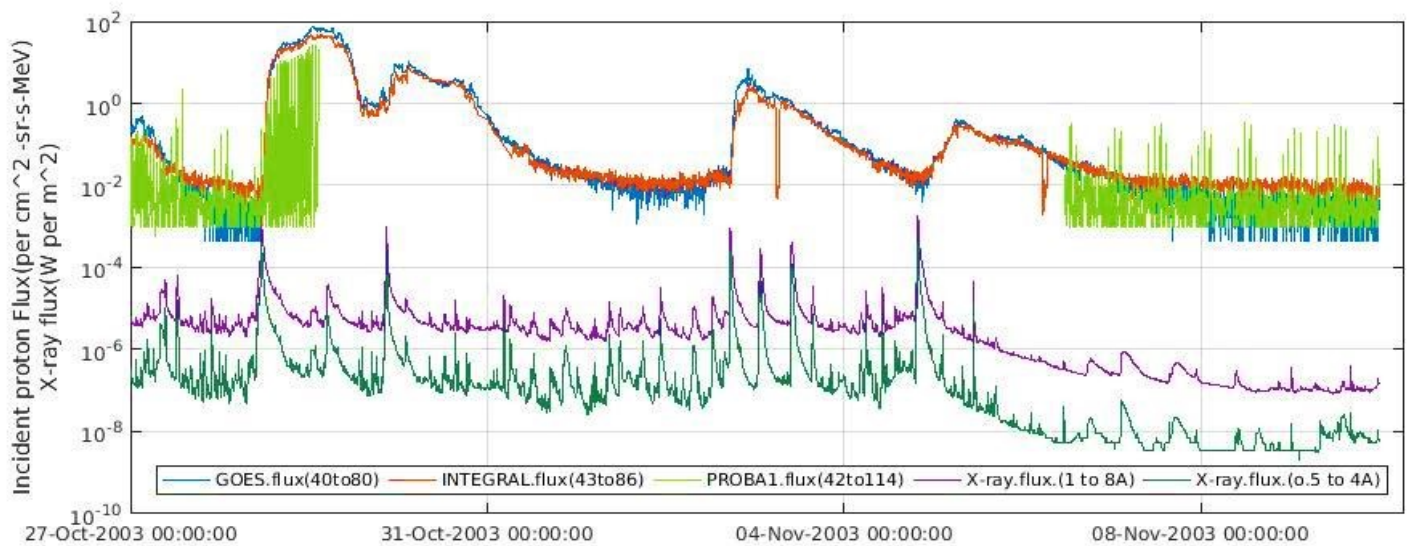


Fig. 10: Incident Proton Fluxes and Solar X-ray flux plotted for Halloween Solar Storm

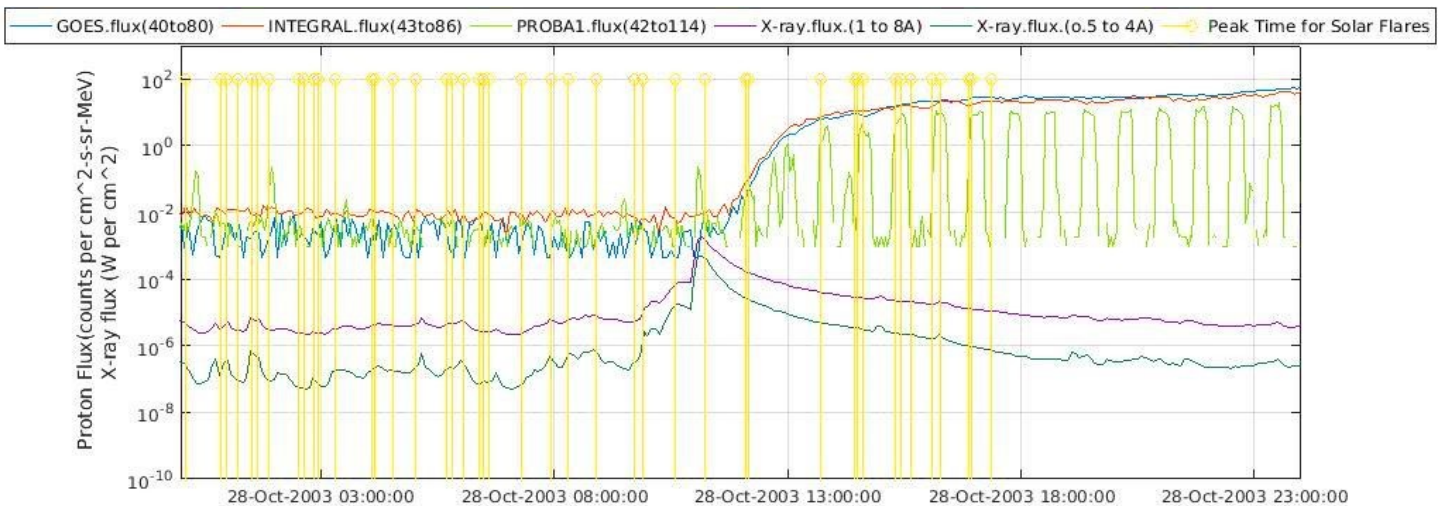


Fig. 11: Proton fluxes, Solar X-ray fluxes & Peak Solar flare time plotted for Halloween solar storm

Next, the trajectories of the three satellites were plotted on the world-map for the same period of Halloween Solar Storm. Three videos were developed using Matlab, each describing the path of the satellite. The screenshots for the same are displayed in Fig. 12, Fig. 13 and Fig. 14 for GOES-10, INTEGRAL and PROBA-1 respectively. Clearly, the three satellites were at different positions with respect to the Earth and yet, the profiles for proton fluxes were matching, pointing in the direction that the whole Earth bathed into the SEPs from the Sun, thus complementing the Halloween Solar Storm.

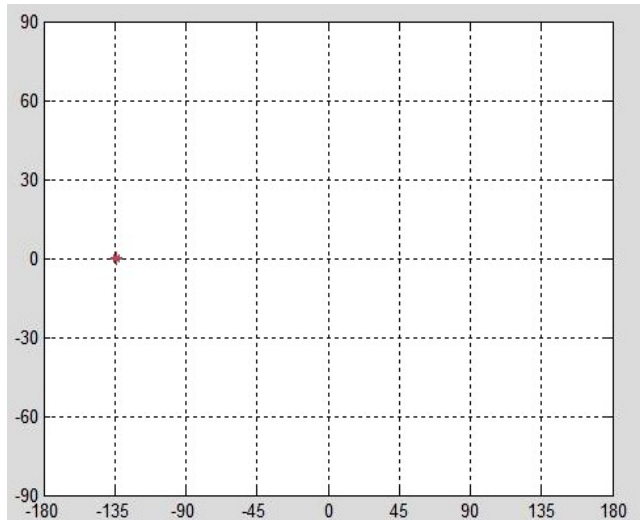


Fig. 12: Trajectory of GOES-10 on a World Map

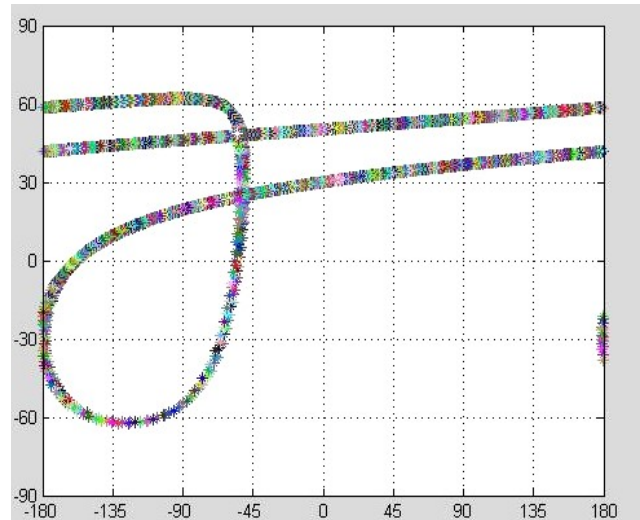


Fig. 13: Trajectory of INTEGRAL on a World Map

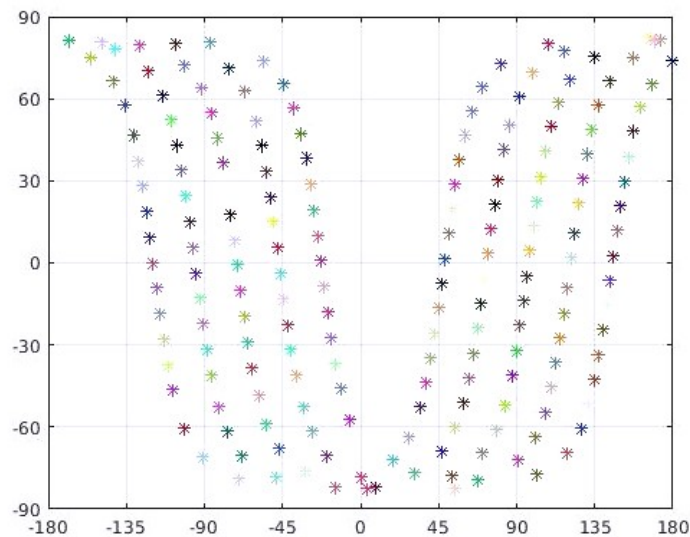


Fig. 14: Trajectory of PROBA-1 on a World Map

CONCLUSION

The report above analyzed the particle environment on the Earth using data from detectors onboard three different satellites IREM- INTEGRAL(ESA), SREM- PROBA(ESA) and SEM- GOES(NASA). The different trajectories of these satellites assisted in inferring proton flux at different locations on the Earth. The period corresponding to the Halloween solar storm (mid Oct to early Nov 2003) was analyzed. The trajectories of the satellites were also plotted on a World map to better co relate the flux with geographical locations. It was observed that the profile of proton flux for all three satellites at different locations complimented each other indicating the whole Earth bathed into SEPs from the Sun.

More advanced tools maybe used to better analyze data and a long-term analysis may provide a way to predict the Space Weather, which may in turn allow raising the alarm signals for people onboard highly flying aircraft, orbiting satellites, astronauts, etc.

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